

The Sangam banks in Prayagraj, Aug. 27



"Carbon emissions from coal, oil and gas, along with deforestation and urbanisation, have supercharged the greenhouse effect, compacting climate change cycles from millennia to a few decades"

— ROXY MATHEW KOLL, *Climate Scientist, Indian Institute of Tropical Meteorology, Pune*



"People must be included in the governance process, be it land-use planning or infrastructure development. There is also a need to decentralise climate data and share it with both citizens and planners"

— MANSHI ASHER, *Co-founder, Himdhara Collective*

evacuation strategies for all extreme weather events. "Currently, India has only one, for cyclones along the coasts," says Ramesh. "There is none for floods. The only strategy used is to put up tents in safe areas, which is more of a reactive measure."

ADAPTATION IS THE KEY

Experts warn that while mitigation softens the effect, adaptation is what will decide survival here and now. Fortunately, adaptation spending rose from 3.7 to 5.6 per cent of GDP between FY16 and FY22. India's cyclone early warning systems have saved thousands of lives, especially in Odisha, but flood, landslide and lightning forecasts are still too broad to act upon. "They must be early enough, and at a hyperlocal level, so that decisions can be made," insists Murtugudde. Lightning, for instance, is a huge emerging risk. Between 2019 and 2025, lightning strikes rose by 400 per cent across the country. In fact, among other climate disasters, lightning consistently caused the highest number of deaths, with figures ranging from 2,357 in 2018 to 2,887 in 2022. Rajasthan has shown the way: after lightning deaths at Amer Fort in 2021, it rolled out village warning systems, built lightning-safe shelters and installed protection devices in schools and hospitals. Fatalities in the state on account of lightning strikes have fallen considerably since.

It is also essential to ensure compliance with climate norms while planning and executing major infrastructure projects. "Any infrastructure development should decrease risk, only then can resilience be built, and future threats, worsened by climate change, be reduced. Infrastructure should not just address present risks but also prepare for emerging ones. That, unfortunately, is not happening," says Ramesh. The consequences are already visible. Several mega infrastructure projects, supposedly resilient, have collapsed in the face of extreme weather. The so-called all-weather highway between Mandi and Manali has seen considerable damage due to floods this year. The Tapovan reservoir in Uttarakhand collapsed in 2021 and, in October 2023, the Teesta III dam in Sikkim was washed away. This must change. Mohanty of IPE Global suggests that climate risk assessment should become a mandatory compliance for all infrastructure projects in addition to environmental compliances.

Any solution must also involve communities. Asher stresses the need to include people in the governance process, be it land-use planning or infrastructure development. She also highlights the need to decentralise climate data and share it with both citizens and planners. Grassroots knowledge about which rivulets flood every year or which slopes slide regularly is invaluable. "Himachal has data at the district level on landslides, but it does not feed into policy for development or urban planning. It does not reach the department of infrastructure, local bodies, or panchayats," says Asher.

The science is clear, the warnings unambiguous and the damage already here. India also has the institutions, the data, and the expertise. What it lacks is consistent enforcement and political will. Added to this are poor compliance with climate norms and communities left out of land-use planning. Regular assessments of emerging climate risks are also ignored. Together, these shortcomings stand in the way of building a truly disaster-resilient India and must be addressed on a war footing. Or else, the country will remain locked in a cycle of rebuilding after each disaster. The culture of *jugaad* cannot patch up a collapsing climate. Prevention, not improvisation, must become the national mindset. The time to act is now, for climate change will not wait for anyone. ■